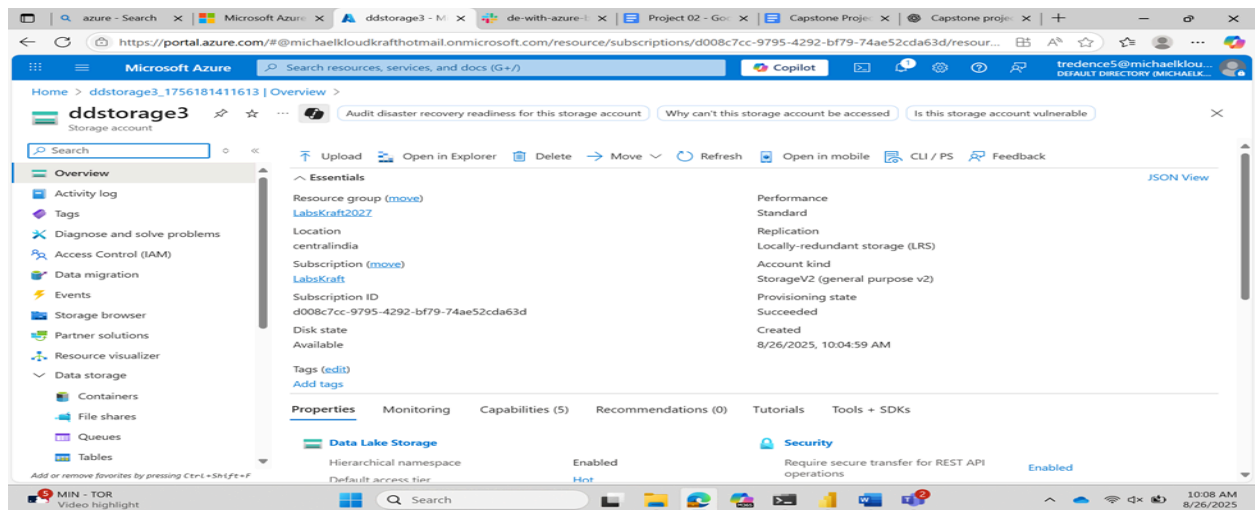
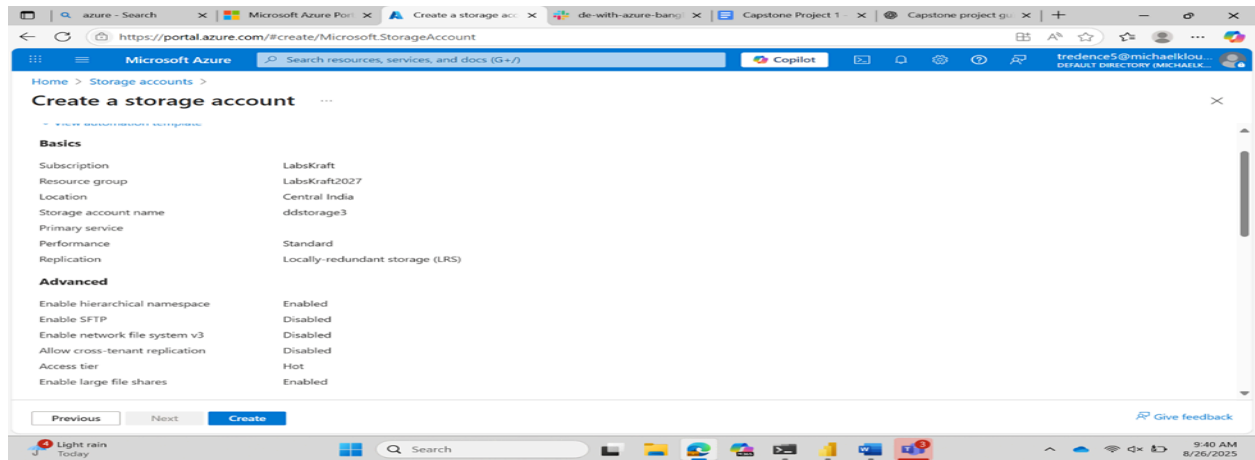


# Devanshu Deo

## Project - 02



azure - Searchddcont3 - MicroCatalog Explore - tred\_d2 - Microproject\_1 - DataAkshita BhamidProject 02 - Go +

https://portal.azure.com/#view/Microsoft\_Azure\_Storage/ContainerMenuBlade/~/overview/storageAccountId/%2Fsubscriptions%2Fd008c7cc-979...

Microsoft AzureSearch resources, services, and docs (G+/)Copilottredence5@michaelklou...DEFAULT DIRECTORY (MICHAELK...

Home > Storage accounts > ddstorage3 | Containers >

ddcont3

Container

Search

+ Add Directory

↑ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔑 Acquire lease

🔑 Break lease

🔧 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

ddcont3

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 2 items

| <input type="checkbox"/> | Name     | Last modified          | Access tier | Blob type | Size | Lease state |
|--------------------------|----------|------------------------|-------------|-----------|------|-------------|
| <input type="checkbox"/> | products | 8/26/2025, 11:37:30 AM |             |           |      | ...         |
| <input type="checkbox"/> | sales    | 8/26/2025, 11:37:42 AM |             |           |      | ...         |

Add or remove favorites by pressing Ctrl+Shift+F

azure - Searchddcont3 - MicroCatalog Explore - tred\_d2 - Microproject\_1 - DataAkshita BhamidProject 02 - Go +Untitled doc +

https://portal.azure.com/#view/Microsoft\_Azure\_Storage/ContainerMenuBlade/~/overview/storageAccountId/%2Fsubscriptions%2Fd008c7cc-979...

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Diagnose and solve problems

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ddcont3 > products

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 2 items

| <input type="checkbox"/> | Name          | Last modified          | Access tier    | Blob type  | Size      | Lease state |
|--------------------------|---------------|------------------------|----------------|------------|-----------|-------------|
| <input type="checkbox"/> | [.]           |                        |                |            |           | ...         |
| <input type="checkbox"/> | products.csv  | 8/26/2025, 11:38:31 AM | Hot (Inferred) | Block blob | 36.69 KiB | Available   |
| <input type="checkbox"/> | products.json | 8/26/2025, 11:38:44 AM | Hot (Inferred) | Block blob | 138.2 KiB | Available   |

Add or remove favorites by pressing Ctrl+Shift+F

azure - Searchddcont3 - MCatalog Expltred\_d2 - Mproject\_1 - T Akshita BhProject 02Untitled doc

https://portal.azure.com/#view/Microsoft\_Azure\_Storage/ContainerMenuBlade/~/overview/storageAccountId/%2Fsubscriptions%2Fd008c7cc-979...

Microsoft AzureSearch resources, services, and docs (G+)Copilottredence5@michaelklou...DEFAULT DIRECTORY (MICHAELK...

Home > ddstorage3 | Containers >

ddcont3

Container

Search

+ Add Directory↑ Upload↻ Refresh🗑 Delete📄 Copy📄 Paste🔄 Rename🔗 Acquire lease🔗 Break lease🔧 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

ddcont3 > sales

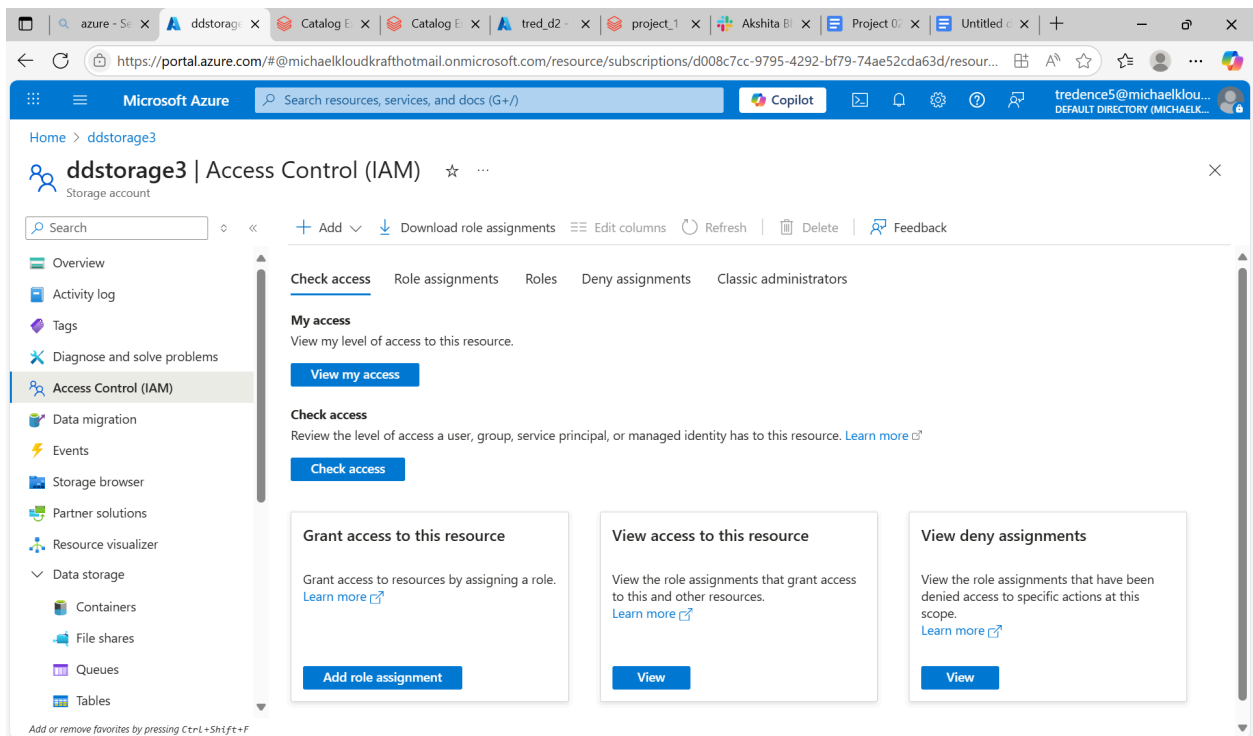
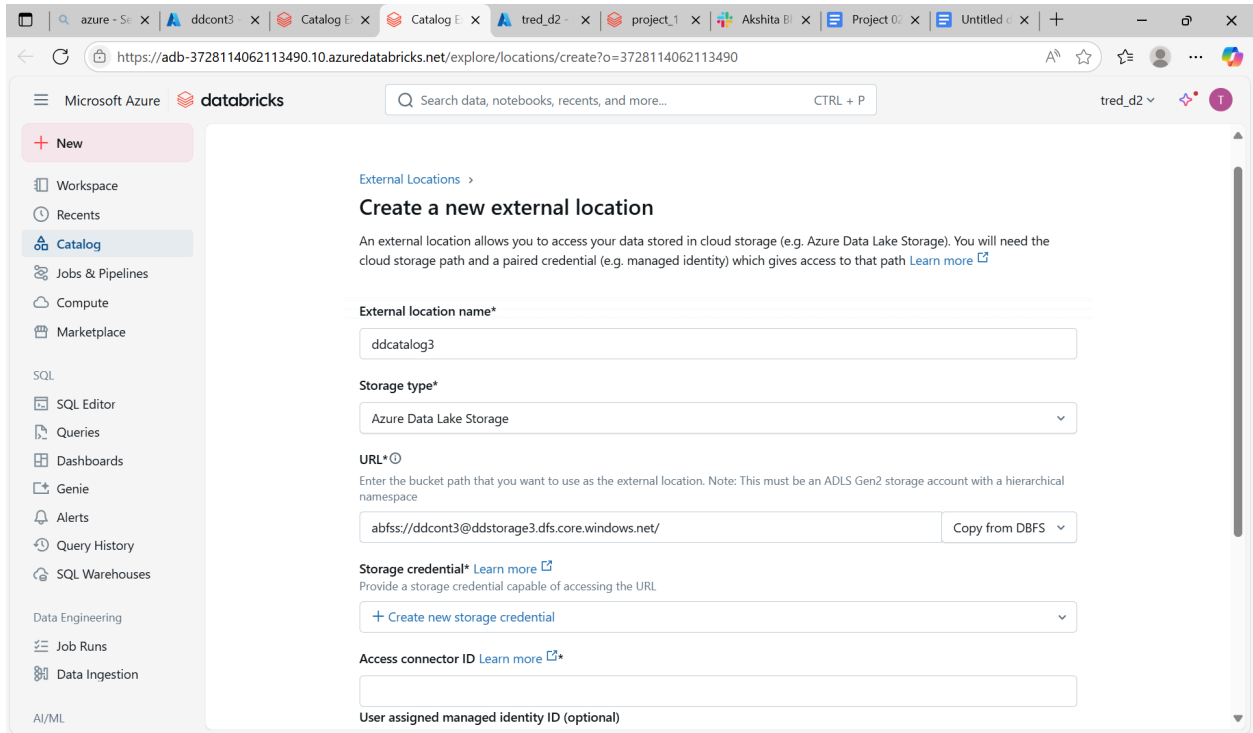
Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)Only show active objects

Showing all 5 items

| <input type="checkbox"/> | Name              | Last modified          | Access tier    | Blob type  | Size      | Lease state |     |
|--------------------------|-------------------|------------------------|----------------|------------|-----------|-------------|-----|
| <input type="checkbox"/> | 📁 [.]             |                        |                |            |           |             | ... |
| <input type="checkbox"/> | 📄 sales_day_1.csv | 8/26/2025, 11:39:51 AM | Hot (Inferred) | Block blob | 50.54 KiB | Available   | ... |
| <input type="checkbox"/> | 📄 sales_day_2.csv | 8/26/2025, 11:40:03 AM | Hot (Inferred) | Block blob | 50.48 KiB | Available   | ... |
| <input type="checkbox"/> | 📄 sales_day_3.csv | 8/26/2025, 11:40:19 AM | Hot (Inferred) | Block blob | 50.46 KiB | Available   | ... |
| <input type="checkbox"/> | 📄 sales_day_4.csv | 8/26/2025, 11:40:33 AM | Hot (Inferred) | Block blob | 50.5 KiB  | Available   | ... |
| <input type="checkbox"/> | 📄 sales_day_5.csv | 8/26/2025, 11:40:48 AM | Hot (Inferred) | Block blob | 50.51 KiB | Available   | ... |

Add or remove favorites by pressing Ctrl+Shift+F



azure - SAdd roleCatalogCatalogtred\_d2project\_1AkshitaBProject 0Untitled

https://portal.azure.com/#view/Microsoft\_Azure\_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2Fd008c7cc-9795-4292-bf79-7...

Microsoft AzureSearch resources, services, and docs (G+)Copilottredence5@michaelklou...DEFAULT DIRECTORY (MICHAELK...

Home > ddstorage3 | Access Control (IAM) >

Add role assignment

RoleMembersConditionsReview + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

**Job function roles**Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Type: AllCategory: All

| Name                               | Description                                                                                    | Type        | Category | Details              |
|------------------------------------|------------------------------------------------------------------------------------------------|-------------|----------|----------------------|
| CreateBlobStorage                  |                                                                                                | CustomRole  | None     | <a href="#">View</a> |
| Defender CSPM Storage Data Scan... | Grants access to read blobs and files. This role is used by the data scanner of Dfender CSPM.  | BuiltInRole | None     | <a href="#">View</a> |
| Defender for Storage Data Scanner  | Grants access to read blobs and update index tags. This role is used by the data scanner of... | BuiltInRole | None     | <a href="#">View</a> |
| Storage Blob Data Contributor      | Allows for read, write and delete access to Azure Storage blob containers and data             | BuiltInRole | Storage  | <a href="#">View</a> |
| Storage Blob Data Owner            | Allows for full access to Azure Storage blob containers and data, including assigning POSI...  | BuiltInRole | Storage  | <a href="#">View</a> |
| Storage Blob Data Reader           | Allows for read access to Azure Storage blob containers and data                               | BuiltInRole | Storage  | <a href="#">View</a> |
| Storage Blob Delegator             | Allows for generation of a user delegation key which can be used to sign SAS tokens            | BuiltInRole | Storage  | <a href="#">View</a> |

Review + assignPreviousNext

Feedback

azure - SAdd roleCatalogCatalogtred\_d2project\_1AkshitaBProject 0Untitled

https://portal.azure.com/#view/Microsoft\_Azure\_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2Fd008c7cc-9795-4292-bf79-7...

Microsoft AzureSearch resources, services, and docs (G+)Copilottredence5@michaelklou...DEFAULT DIRECTORY (MICHAELK...

Home > ddstorage3 | Access Control (IAM) >

Add role assignment

RoleMembersConditionsReview + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)

[View my access](#)

Selected role

Storage Blob Data Contributor

Assign access to

User, group, or service principal

☒ Managed identity

Members

+ Select members

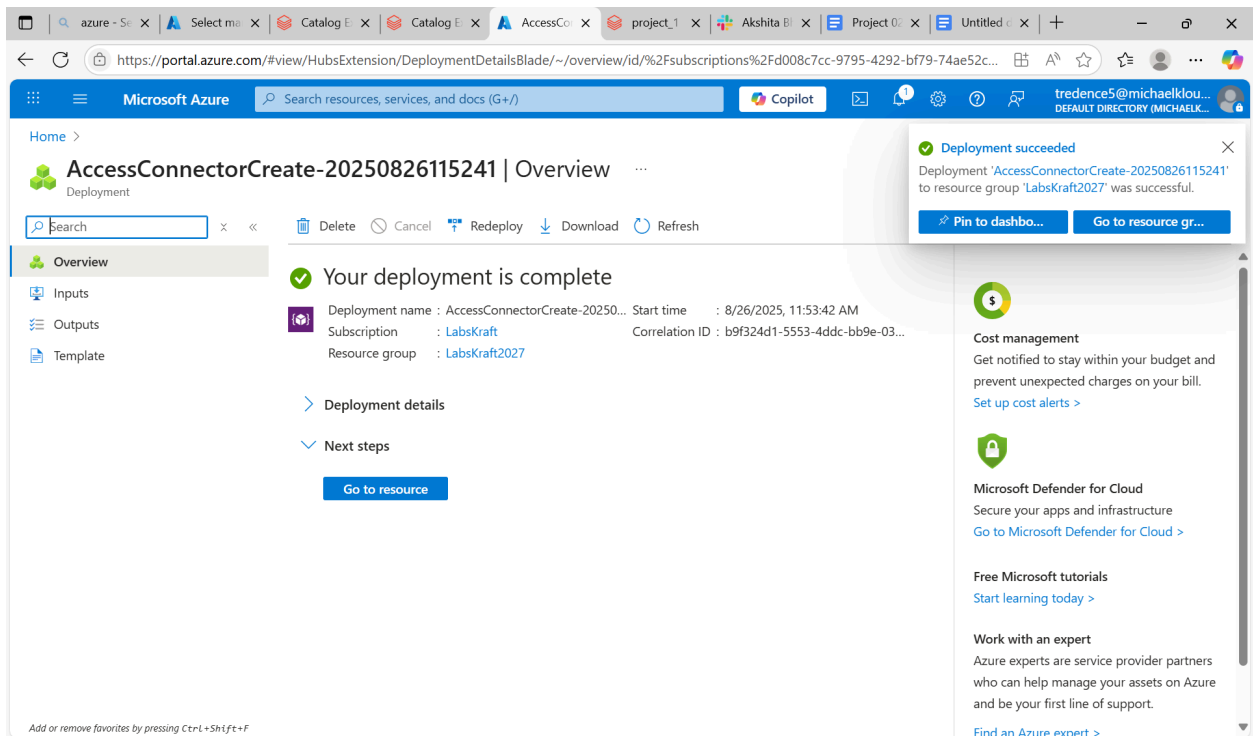
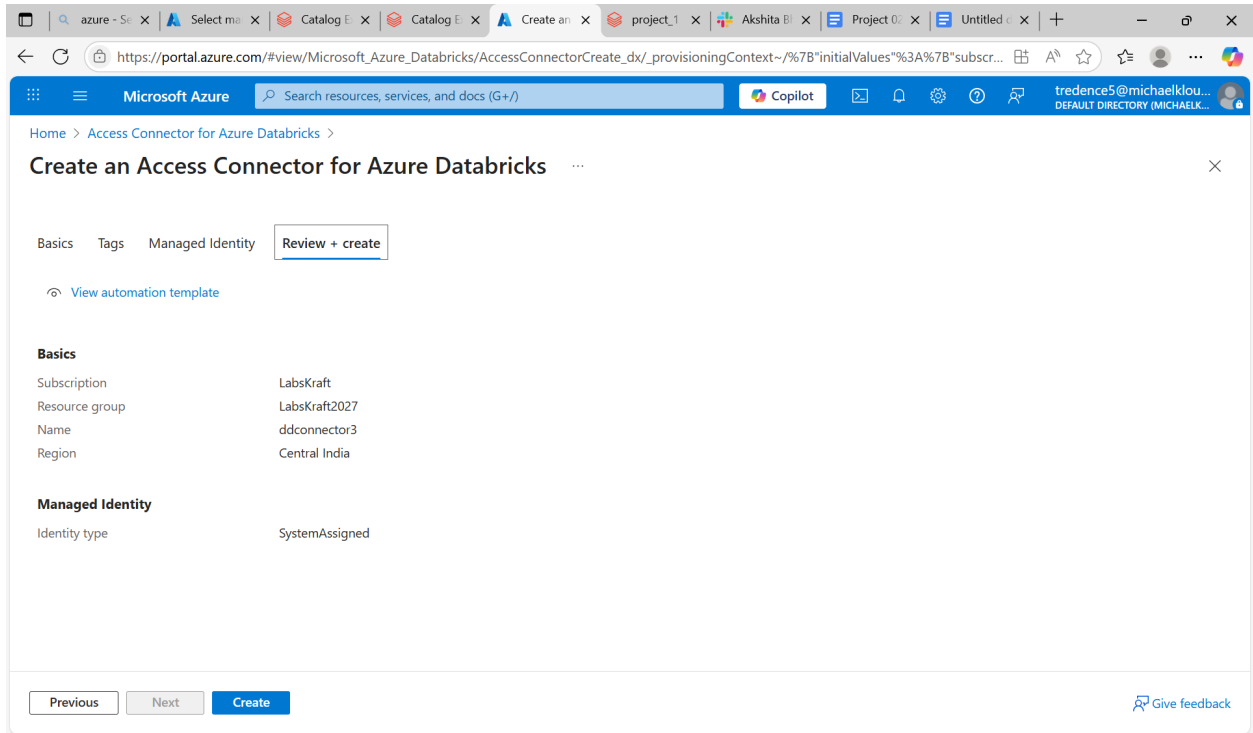
| Name                | Object ID | Type |
|---------------------|-----------|------|
| No members selected |           |      |

Description

Optional

Review + assignPreviousNext

Feedback



azure - S...Select m...Catalog E...Catalog E...ddconnec...project\_1...Akshita B...Project 0...Untitled (...+--

https://portal.azure.com/#view/Microsoft\_Azure\_AD/AddRoleAssignmentsLandingBlade/scope/%2Fsubscriptions%2Fd008c7cc-9795-4292-bf79-7...

Microsoft AzureSearch resources, services, and docs (G+/)Copilottredence5@michaelklou...DEFAULT DIRECTORY (MICHAELK...

Home > ddstorage3 | Access Control (IAM) >

Add role assignment

RoleMembersConditionsReview + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)  
[View my access](#)

Selected role

Storage Blob Data Contributor

Assign access to

User, group, or service principal

Managed identity

Members

+ Select members

| Name                | Object ID |
|---------------------|-----------|
| No members selected |           |

Description

Optional

Review + assign

Previous

Next

Select managed identities

Some results might be hidden due to your ABAC condition.

Subscription \*

LabsKraft

Managed identity

Access Connector for Azure Databricks (24)

dd

Selected members:

 ddconnector3  
/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/... [Remove](#)

Select

Close

Feedback

azure - S...ddstorag...Catalog E...Catalog E...ddconnec...project\_1...Akshita B...Project 0...Untitled...+

←↻https://portal.azure.com/#@michaelkloudkrafthotmail.onmicrosoft.com/resource/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resour...A☆🔍👤...

Microsoft Azure 🔍 Search resources, services, and docs (G+) Copilot 📧🔔⚙️🕒💬👤tredence5@michaelklou...DEFAULT DIRECTORY (MICHAELK...

Home > ddstorage3

ddstorage3 | Access Control (IAM) ☆...

Storage account

🔍 Search 🔍

⏏️

+ Add ▾

📄 Download role assignments

⌵ Edit columns

🔄 Refresh

🗑️ Delete

🗣️ Feedback

Overview

Activity log

Tags

⚡ Diagnose and solve problems

Access Control (IAM)

📦 Data migration

⚡ Events

📁 Storage browser

🤖 Partner solutions

📊 Resource visualizer

▼ Data storage

📁 Containers

📁 File shares

📁 Queues

📁 Tables ☆

Added Role assignment

ddconnector3 was added as Storage Blob Data Contributor for ddstorage3.

×

Check accessRole assignmentsRolesDeny assignmentsClassic administrators

My access

View my level of access to this resource.

View my access

Check access

Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

Check access

Grant access to this resource

Grant access to resources by assigning a role. [Learn more](#)

Add role assignment

View access to this resource

View the role assignments that grant access to this and other resources. [Learn more](#)

View

View deny assignments

View the role assignments that have been denied access to specific actions at this scope. [Learn more](#)

View

https://portal.azure.com/#@michaelkloudkrafthotmail.onmicrosoft.com/resource/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/LabsKraft2027/providers/Microsoft.Storage/storageAccounts/ddstorage3/ta...



Microsoft Azure databricks

Search data, notebooks, recents, and more...

+ New

- Workspace
- Recents
- Catalog
- Jobs & Pipelines
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion

AI/ML

Catalog Explorer > External Locations >

### ddcatalog3

Overview Permissions Browse Workspaces

Description

Add description

|                        |                                                  |
|------------------------|--------------------------------------------------|
| Credential             | ddcatalog3_azuremanagedidentity_1756189653552    |
| URL                    | abfss://ddcont3@ddstorage3.dfs.core.windows.net/ |
| Limit to read-only use | Disabled                                         |

About this external location

Owner Tredence LabsKraft5

Fallback mode: ☐

Successful!

Your external location: ddcatalog3 has been created

Test connection

Microsoft Azure databricks

Search data, notebooks, recents, and more...

+ New

- Workspace
- Recents
- Catalog
- Jobs & Pipelines
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion

AI/ML

### Create a new catalog

A catalog is the first layer of Unity Catalog's three-level namespace and is used to organize your data assets. [Learn more](#)

Catalog name\*

dd\_catalog33

Type\*

Standard

Storage location

ddcatalog3 sub/path

[Create a new external location](#)

abfss://ddcont3@ddstorage3.dfs.core.windows.net/

Location in cloud storage where data for managed tables will be stored. If not specified, the location will default to the metastore root location.

Cancel Create

dd data

Filter

| Last viewed | Type    |
|-------------|---------|
| 1 day ago   | Table   |
| 1 day ago   | Schema  |
| 1 day ago   | Catalog |
| 1 day ago   | Schema  |
| 3 days ago  | Schema  |
| 3 days ago  | Table   |
| 3 days ago  | Catalog |
| 5 days ago  | Catalog |

azure - SddconnectCatalog Eddcatalogddconnectproject\_1Akshita BProject 0Untitled

https://adb-3728114062113490.10.azuredatabricks.net/explore/data?o=3728114062113490

Microsoft Azure databricks

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tred\_d2

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Catalog

Serverless Starter WarehouseServerless

Type to search...

My organization

tred\_d2

system

dd\_cata

dd\_cata

dd\_cata

ddcata

tredenc

aman\_c

capston

diksha\_capstone\_catalog

diksha\_second\_capstone

diksha-first-catalog

diksha-second-catalog

haseebcat

haseebscatalog

hetul\_catalog

hetul\_new\_catalog

hetul\_project01\_catalog

Delta Sharing

Clean Rooms

External Data

Add data

Send feedback

Quick access

Recents

Favorites

Catalogs

Filter

Catalog created!

Catalog has been created with recommended permission settings. Click configure to set up additional permissions, workspace bindings, and to add metadata.

View catalog

Configure catalog

Last viewed

Type

1 day ago

Table

1 day ago

Schema

1 day ago

Catalog

1 day ago

Schema

3 days ago

Schema

3 days ago

Table

3 days ago

Catalog

5 days ago

Catalog

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

tred\_d2

**dd\_catalog33**

Overview Details Permissions Workspaces

Description

AI generate Add

Filter schemas 2 schemas

| Name               | Owner                            | Created at             |
|--------------------|----------------------------------|------------------------|
| default            | tredence5@michaelkloudkraftth... | Aug 26, 2025, 12:01 PM |
| information_schema | System user                      | Aug 26, 2025, 12:01 PM |

About this catalog

Owner Tredence LabsKraft5

Tags

Add tags

My organization

- tred\_d2
- system
- dd\_catalog
- dd\_catalog33
  - default
  - information\_schema
- dd\_catalog77
- ddcatalog5
- tredence\_02
- aman\_cata
- capstone03
- diksha\_capstone\_catalog
- diksha\_second\_capstone
- diksha-first-catalog
- diksha-second-catalog
- haseebcat
- haseebscatalog
- hatul catalog

Grant on dd\_catalog33

Granted privileges will be inherited by applicable objects (e.g. schemas, tables) in this catalog. [Learn more](#)

Principals

Type to add multiple principals

Privilege presets

Custom

Prerequisite

- ☐ USE CATALOG
- ☐ USE SCHEMA

Metadata

- ☐ APPLY TAG
- ☐ BROWSE

Read

- ☐ EXECUTE
- ☐ READ VOLUME
- ☐ SELECT

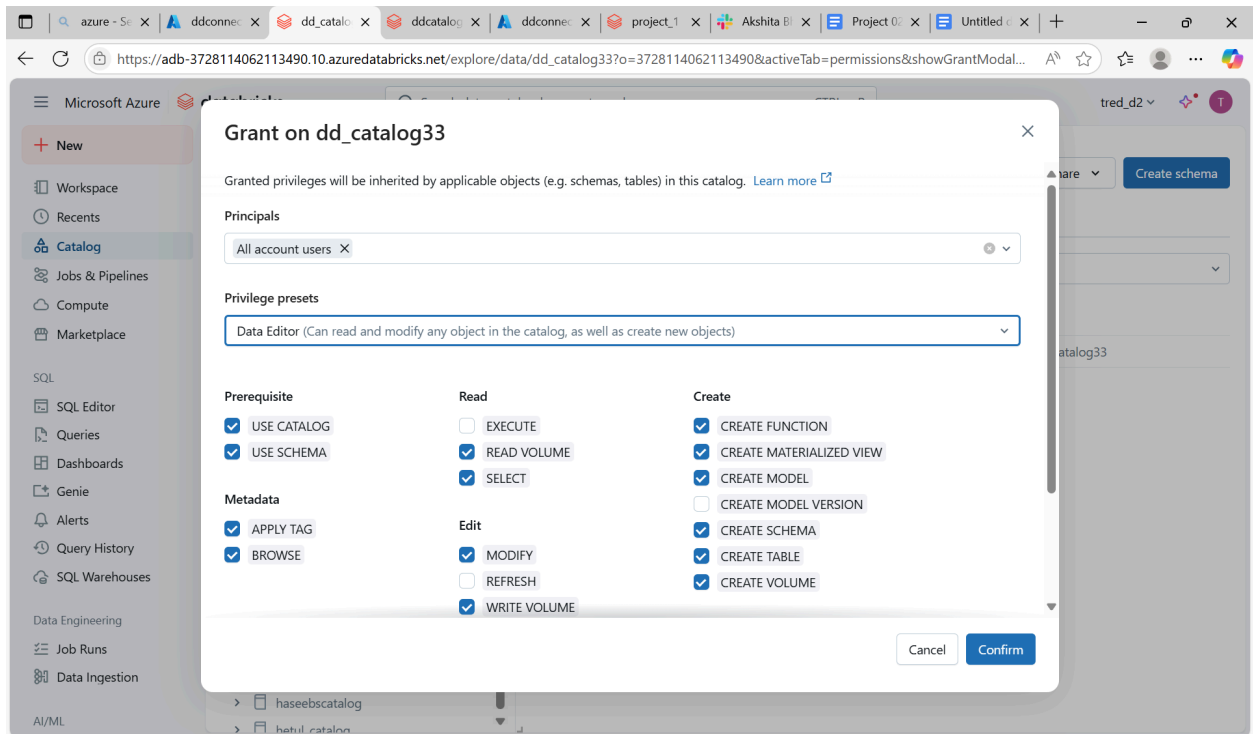
Edit

- ☐ MODIFY
- ☐ REFRESH
- ☐ WRITE VOLUME

Create

- ☐ CREATE FUNCTION
- ☐ CREATE MATERIALIZED VIEW
- ☐ CREATE MODEL
- ☐ CREATE MODEL VERSION
- ☐ CREATE SCHEMA
- ☐ CREATE TABLE
- ☐ CREATE VOLUME

Cancel Confirm



```
%sql
USE CATALOG dd_catalog33;
create schema if not exists dd_catalog33.sales_analytics;
```

```
import dlt
from pyspark.sql.functions import *

# Ingest raw sales data (multiple daily CSV files in raw/sales/)
@dlt.table(
    name="sales_raw",
    comment="Raw sales data ingested from CSV files",
    table_properties={"quality": "bronze"}
)
```

```

def sales_raw():
    return (
        spark.readStream.format("cloudFiles") # Auto Loader for incremental
loads
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .load("abfss://ddcont3@ddstorage3.dfs.core.windows.net/sales/")
    )

# Products CSV
@dlt.table(
    name="products_raw_csv",
    comment="Raw product data from CSV",
    table_properties={"quality": "bronze"}
)
def products_raw_csv():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .load("abfss://ddcont3@ddstorage3.dfs.core.windows.net/products/")
    )

# Products JSON
@dlt.table(
    name="products_raw_json",
    comment="Raw product data from JSON",
    table_properties={"quality": "bronze"}
)
def products_raw_json():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "json")
        .load("abfss://ddcont3@ddstorage3.dfs.core.windows.net/products/")
    )

# Unified Products (CSV + JSON)
@dlt.view(name="products_raw")

```

```

def products_raw():
    return

dlt.read("products_raw_csv").unionByName(dlt.read("products_raw_json"),
allowMissingColumns=True)

# Clean Sales
@dlt.table(
    name="sales_clean_table",
    comment="Cleaned sales data with enforced schema",
    table_properties={"quality": "silver"}
)
@dlt.expect("valid_sales_amount", "sales_amount >= 0")
@dlt.expect("valid_quantity", "quantity > 0")
def sales_clean_table():
    return (
        dlt.read("sales_raw")
        .withColumn("sale_id", col("sale_id").cast("string"))
        .withColumn("product_id", col("product_id").cast("string"))
        .withColumn("region", col("region").cast("string"))
        .withColumn("store_id", col("store_id").cast("string"))
        .withColumn("quantity", col("quantity").cast("int"))
        .withColumn("sales_amount", col("sales_amount").cast("double"))
        .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
        .na.drop()
    )

@dlt.table(
    name="products_clean",
    comment="Cleaned products data with schema evolution support",
    table_properties={"quality": "silver"}
)
def products_clean():
    return (
        dlt.read("products_raw")
        .withColumn("product_id", col("product_id").cast("string"))
        .withColumn("product_name", col("product_name").cast("string"))
        .withColumn("category", col("category").cast("string"))
        .withColumn("price", col("price").cast("double"))
    )

```

```

        .withColumn("discount_rate", col("discount_rate").cast("double"))
    )

# Join sales + products and calculate KPIs
@dlt.table(
    name="sales_summary",
    comment="Daily sales summary by product and region",
    table_properties={"quality": "gold"}
)

def sales_summary():
    sales = dlt.read("sales_clean_table")
    products = dlt.read("products_clean")
    return (
        sales.join(products, "product_id", "left")
        .groupBy("region", "sale_date", "product_id", "product_name",
"category")
        .agg(
            sum("sales_amount").alias("daily_revenue"),
            sum("quantity").alias("units_sold"),
            avg("price").alias("avg_price"),
            avg("discount_rate").alias("avg_discount")
        )
    )

```

Microsoft Azure databricks

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Jobs & Pipelines >

### Create pipeline [Send feedback](#)

**General**

\* Pipeline name

dd\_pipeline3

☐ Serverless

Product edition

Advanced

Core

Pro

Advanced

Run as

tredence5@michaelkloudkrafthotmail.onmicrosoft.com

This user must have permissions to the notebooks and source datasets, as well as target catalog and

Cancel Create

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

Jobs & Pipelines >

### dd\_pipeline3 [Send feedback](#)

Run: 8/26/2025, 12:32:04 PM · Created...

Waiting for resources

3 Initializing

4 Setting up tables

**Pipeline details** Update details

Pipeline ID a44ac3f5-3c5c-4f2c-864e-ed9d229d2ff7

Pipeline type ETL pipeline

Source code /Users/tredence5@michaelkloudkrafthotmail.onmicrosoft.com/Project\_2

**Event log** Query history

All Info Warning Error Filter...

11 seconds ago user\_action User tredence5@michaelkloudkrafthotmail.onmicrosoft.com started an update.

11 seconds ago create\_update Update fc13ff started by USER\_ACTION.

10 seconds ago update\_progress Update fc13ff is WAITING\_FOR\_RESOURCES.



Microsoft Azure databricks

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Jobs & Pipelines >

dd\_pipeline3 ☆ Send feedback

Run: 8/26/2025, 4:09:18 PM · Completed Select tables for refresh

Graph List

Show side panel

Event log Query history

All Info Warning Error Filter...

3 minutes ago flow\_progress Flow 'dd\_catalog33.sales\_analytics.sales\_summary' is RUNNING.

25 seconds ago dataset\_life\_cycle Materialized View 'dd\_catalog33'.sales\_analytics'.sales\_summary' has been created.

25 seconds ago flow\_progress Flow 'dd\_catalog33.sales\_analytics.sales\_summary' has COMPLETED.

17 seconds ago update\_progress Update 3487e9 is COMPLETED.

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

Jobs & Pipelines >

dd\_pipeline3 ☆ Send feedback

Run: 8/26/2025, 4:28:26 PM · Canceled This update has already finished. Show the latest update

Graph List

Event log Query history

All Info Warning Error Filter...

2 minutes ago flow\_progress Flow 'dd\_catalog33.sales\_analytics.products\_clean' is STARTING.

2 minutes ago flow\_progress Flow 'dd\_catalog33.sales\_analytics.products\_clean' is RUNNING.

1 minute ago flow\_progress Flow 'dd\_catalog33.sales\_analytics.products\_clean' has COMPLETED.

1 minute ago update\_progress Update 316e0e has been cancelled due to a schema change in dd\_catalog33.sales\_analytics.sales\_raw, and will be r...

```

import dlt
from pyspark.sql.functions import *
from pyspark.sql.types import *

# Configuration for schema locations - these are still needed to track schema
# evolution
sales_schema_location =
"abfss://ddcont3@ddstorage3.dfs.core.windows.net/schema/sales/"
products_csv_schema_location =
"abfss://ddcont3@ddstorage3.dfs.core.windows.net/schema/products_csv/"
products_json_schema_location =
"abfss://ddcont3@ddstorage3.dfs.core.windows.net/schema/products_json/"

# Note: We no longer define explicit schemas here.
# The pipeline will now infer the schema from the data itself.

# Ingest raw sales data with automated schema evolution
@dlt.table(
    name="sales_raw",
    comment="Raw sales data ingested from CSV files with automated schema
inference",
    table_properties={"quality": "bronze"}
)
def sales_raw():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .option("cloudFiles.schemaLocation", sales_schema_location)
        .option("cloudFiles.schemaEvolutionMode", "addNewColumns") #
Automatically adds new columns
        .load("abfss://ddcont3@ddstorage3.dfs.core.windows.net/sales/")
    )

# Products CSV with automated schema evolution
@dlt.table(
    name="products_raw_csv",

```

```

        comment="Raw product data from CSV with automated schema inference",
        table_properties={"quality": "bronze"}
    )
def products_raw_csv():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .option("header", "true")
        .option("cloudFiles.schemaLocation", products_csv_schema_location)
        .option("cloudFiles.schemaEvolutionMode", "addNewColumns")
        .load("abfss://ddcont3@ddstorage3.dfs.core.windows.net/products/")
    )

# Products JSON with automated schema evolution
@dlt.table(
    name="products_raw_json",
    comment="Raw product data from JSON with automated schema inference",
    table_properties={"quality": "bronze"}
)
def products_raw_json():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "json")
        .option("multiLine", "true") # Crucial for multi-line JSON files
        .option("cloudFiles.schemaLocation", products_json_schema_location)
        .option("cloudFiles.schemaEvolutionMode", "addNewColumns")
        .load("abfss://ddcont3@ddstorage3.dfs.core.windows.net/products/")
    )

# Unified Products (CSV + JSON)
@dlt.view(name="products_raw")
def products_raw():
    return

dlt.read_stream("products_raw_csv").unionByName(dlt.read_stream("products_raw_json"), allowMissingColumns=True)

# Clean Sales with data quality rules. We rely on inferred schema.
@dlt.table(

```

```

    name="sales_clean_table",
    comment="Cleaned sales data with data quality checks",
    table_properties={"quality": "silver"}
)
@dlt.expect_or_drop("valid_sales_amount", "sales_amount >= 0")
@dlt.expect_or_drop("valid_quantity", "quantity > 0")
def sales_clean_table():
    # We now read the stream directly and apply transformations.
    # The schema is already inferred from the source.
    return (
        dlt.read_stream("sales_raw")
        .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
        .na.drop()
    )

# Clean Products. We rely on inferred schema and coalesce on discount_rate.
@dlt.table(
    name="products_clean_table",
    comment="Cleaned products data with schema evolution support",
    table_properties={"quality": "silver"}
)
def products_clean_table():
    return (
        dlt.read_stream("products_raw")
        .withColumn("discount_rate", coalesce(col("discount_rate"), lit(0.0)))
    )

# Join sales + products and calculate KPIs
@dlt.table(
    name="sales_summary",
    comment="Daily sales summary by product and region",
    table_properties={"quality": "gold"}
)
def sales_summary():
    sales = dlt.read_stream("sales_clean_table")
    products = dlt.read("products_clean_table")
    return (
        sales.join(products, "product_id", "left")
    )

```

```

.groupBy("region", "sale_date", "product_id", "product_name",
"category")
.agg(
    sum("sales_amount").alias("daily_revenue"),
    sum("quantity").alias("units_sold"),
    avg("price").alias("avg_price"),
    avg("discount_rate").alias("avg_discount")
)
)
)

```

The screenshot displays the Databricks workspace interface. The top navigation bar includes the Microsoft Azure logo, the Databricks logo, and a search bar. The left sidebar contains a 'New' button and a list of workspace components: Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, and SQL Warehouses. The main area shows a notebook titled 'Project\_2' with two tabs: 'Python' and 'SQL'. The 'Python' tab is active, showing a notebook with the following code:

```

USE CATALOG dd_catalog33;
create schema if not exists dd_catalog33.sales_analytics;

```

Below the code editor, the 'DLT graph' is visible, showing a data pipeline. The graph includes the following components:

- products\_raw\_csv**: Streaming table, Completed - 24s.
- products\_raw\_json**: Streaming table, Completed - 21s.
- products\_raw**: View, Completed - 10s.
- products\_clean**: Materialized view, Completed - 24s.
- sales\_raw**: Streaming table, Completed - 59s.
- sales\_clean\_table**: Materialized view, Completed - 13s.
- sales\_summary**: Materialized view, Completed - 30s.

The graph shows the flow of data from the raw tables to the cleaned tables and finally to the summary table.

azure | ddcon | dd\_pip | dd\_pip | Cc X | ddcata | ddcon | Jobs - | Akshite | Project | Untitle | New ta | keybr | Practic | +

https://adb-3728114062113490.10.azuredatabricks.net/compute/sql-warehouses?o=3728114062113490&page=1&page\_size=20

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

tred\_d2

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Compute

All-purpose compute Job compute SQL warehouses Vector Search Pools Policies Apps OLTP Database

Filter SQL warehouses

Only my SQL warehouses

Created by

Size

Status

Type

Create SQL warehouse

| Status | Name                         | Created by | Size  | Active / Max | Type       |
|--------|------------------------------|------------|-------|--------------|------------|
|        | Serverless Starter Warehouse | Michael A  | Small | 1 / 1        | Serverless |

1

Get Data

Search

All

File

Database

Microsoft Fabric

Power Platform

Azure

Online Services

Other

Azure

 Azure SQL database

 Azure Synapse Analytics SQL

 Azure Analysis Services database

 Azure Database for PostgreSQL

 Azure Blob Storage

 Azure Table Storage

 Azure Cosmos DB v1

 Azure Data Explorer (Kusto)

 Azure Data Lake Storage Gen2

 Azure HDInsight (HDFS)

 Azure HDInsight Spark

 HDInsight Interactive Query

 Azure Cost Management

 Azure Resource Graph

 Azure Cosmos DB v2

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Template Apps

Connect

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SQL Warehouses >

### Serverless Starter Warehouse

Overview **Connection details** Monitoring

Use these details to connect to this warehouse [Create a personal access token](#)

Tableau Power BI dbt Python Java Node.js Go More tools

**Server hostname**  
adb-3728114062113490.10.azuredatabricks.net

**HTTP path**  
/sql/1.0/warehouses/3f8c64908dc08658

**JDBC URL** 2.6.25 or later

```
jdbc:databricks://adb-3728114062113490.10.azuredatabricks.net:443/default;transportMode=http;ssl=1;AuthMech=3;httpPath=/sql/1.0/warehouses/3f8c64908dc08658;
```

Databricks supports drivers released within the last two years. [Download drivers here](#)

